
Chemical Specificity and Conformational Limits of Hierarchical Uncertainty in HIP-NN

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Abstract

An energy hierarchy can correlate with prediction error while providing an incomplete warning of failure. We audit this distinction in HIP-NN, holding its architecture and training protocol fixed and extrapolating from QM9 molecules with at most eight heavy atoms to molecules with nine. Three converged fits achieve an independent smaller-molecule MAE of 0.317 ± 0.007 kcal/mol, increasing to 0.780 ± 0.004 on larger molecules. Seven of twelve predefined motifs show joint error–hierarchy rank associations in every seed after family-wise correction. Their coverage is chemically broad: heterocycles alone contain $85.4 \pm 0.2\%$ of failures but also 71.7% of evaluation molecules. A Continuation and Cancellation Audit augments the original hierarchy ratio with a bounded dimensional continuation signal. Its held-out calibration error is 0.030 ± 0.004 kcal/mol, with all ten bin means within 10% of observed error in every seed; nevertheless, recall at the fixed 1 kcal/mol warning threshold is only $33.2 \pm 2.0\%$. On 9,244 held-out ANI-1x conformers, four motif associations persist across seeds, while relative-energy discrepancy reaches 19.90 ± 0.15 kcal/mol. The results identify reproducible chemical structure in hierarchy failure, but distinguish vocabulary compactness from specificity and mean-error calibration from reliable protection against individual or conformational failures.

1 A hierarchy can diagnose failure without measuring its cost

Accurate average molecular energies do not establish that an interatomic potential can identify its own unreliable predictions. In the original HIP-NN study, a random-split QM9 MAE of 0.256 ± 0.003 kcal mol⁻¹ coexisted with $2.26 \pm 0.07\%$ of errors exceeding 1 kcal mol⁻¹ [Lubbers et al., 2018, Tables I and II]. Such tails matter when screening selects a small number of apparently favorable molecules or when errors accumulate along a trajectory. A useful diagnostic must distinguish unusually risky chemistry, attach a meaningful energy scale to that risk, and survive the change from optimized molecules to non-equilibrium conformations. These are different requirements.

HIP-NN offers an unusually direct starting point. Its energy is a sum of atomwise contributions at successive interaction orders, and training penalizes contributions that fail to decay. The non-hierarchicality R correlates with error quantiles in the original random-split experiment [Lubbers et al., 2018, Fig. 3]. That correlation leaves several questions open. Does a large score identify particular chemical environments under molecular-size extrapolation? Do a few such environments cover the error tail, or does coverage merely reflect common substructures? Can the hierarchy predict the *magnitude* of error, and are the chemical associations preserved away from equilibrium?

We study these questions while holding the energy network fixed. Models learn only from QM9 molecules with at most eight heavy atoms; nine-heavy-atom molecules supply distinct diagnostic-fitting, calibration, and evaluation partitions. We test a finite, outcome-independent family of structural motifs, retaining null associations and correcting the entire confirmatory family. We also examine a dimensional continuation score constructed from the existing hierarchy. Its calibration is an ordinary post-hoc regression, not a new probabilistic potential. Finally, an independently sampled

ANI-1x panel tests the chemical library using within-molecule relative energies. This design separates prediction quality, chemical interpretability, and conformational portability instead of treating any one as evidence for the others.

Relation to uncertainty estimation. Ensembles estimate disagreement between fitted potentials, but disagreement can understate out-of-distribution error and requires system-specific calibration [Kahle and Zipoli, 2022]. Dropout potentials [Wen and Tadmor, 2020] and variational Bayesian MLIPs [Coscia et al., 2026] introduce parameter uncertainty; fast single-network approaches reduce the cost of uncertainty evaluation [Zhu et al., 2023]. Conformal methods calibrate latent-distance or other scores into intervals under exchangeability [Hu et al., 2022]; more recent environment-dependent calibration learns adaptive quantile functions [Ho et al., 2026]. Our target is conditional *mean absolute error*, so none of our calibration claims implies interval coverage.

Chemical interpretation itself is also not new. Vazquez-Salazar et al. [2022] examined chemically structured failures and uncertainty–error mismatches in evidential PhysNet predictions, including out-of-distribution tautomeric environments. Our study concerns a different object: the order-resolved decomposition of an unchanged, hierarchically regularized potential. The contribution is the joint error–hierarchy audit, its explicit coverage-versus-specificity accounting, and its conformational transfer test. Neither generic isotonic calibration nor the familiar chemical substructures are claimed as independently new methods.

2 An outcome-separated chemical audit

All candidate definitions, signal formulas, splits, and thresholds were recorded before test predictions were inspected. A timestamped addendum specifies connectivity-level inference and discovery-only selection for compactness. The complete protocol, source hashes, assignments, and numerical outputs accompany the manuscript.

2.1 Data and the energy predictor

We use the staged QM9 release [Ramakrishnan et al., 2014], removing its 3,054 geometric-consistency failures and the eleven difficult minimizations listed in the release documentation. Three minimization cases already occur in the consistency list, leaving 130,823 molecules. Of these, 21,799 contain at most eight heavy atoms and 109,024 contain exactly nine; heavy atoms are C, N, O, or F. The target is the zero-temperature energy U_0 , expressed as an atomization energy by subtracting the release’s isolated-atom U_0 values. This subtraction is computed from the supplied reference file. It fixes the energy convention for the hierarchy and does not change molecular prediction errors.

A fixed random sample of 2,000 smaller molecules is reserved as an independent in-distribution (ID) audit. Seeds 11, 22, and 33 independently shuffle the remaining smaller molecules into 18,799 optimizer-training and 1,000 validation examples. The extra ID audit is a deliberate departure from assigning every non-validation smaller molecule to training: early-stopping validation error is not an independent reproduction metric. Nine-heavy-atom molecules never train the energy network. Canonical connectivity graphs, with stereochemistry removed, are randomly divided into 40% diagnostic fitting, 20% calibration, and 40% final evaluation, using RNG seed 1729. This yields 43,609, 21,799, and 43,616 molecules, respectively, with no connectivity shared across these partitions. Each model is nevertheless evaluated and logged on *every* nine-heavy-atom molecule. The final partition alone supports calibration and confirmatory motif claims.

We instantiate the staged hippynn implementation [LANL hippynn contributors, 2026]: two interaction blocks, three on-site layers after each interaction, 80 features, 20 learned inverse-distance sensitivities, softplus activations, residual connections, and the source-paper cosine cutoff of 15 Bohr. Sensitivity centers initially span 1.7–10 Bohr. There are 234,327 parameters, including five frozen atomic-baseline coefficients. A least-squares fit to optimizer-training energies fixes the zeroth-order readout. Uniform Glorot initialization and zero biases replace the package’s default initialization. Higher readouts start at dimensionless scales 10^{-2n} , following the paper’s stated decay intention; the printed denominator 10^{-2n} in its initialization paragraph would instead imply growth. This explicit interpretation and standardizing the readout coefficients are implementation choices.

Let σ_E be the training atomization-energy standard deviation. The loss is

$$\mathcal{L} = \frac{\text{MAE} + \text{RMSE}}{\sigma_E} + 10^{-6} \|W_{\text{reg}}\|_2^2 + 10^{-2} \langle R \rangle, \quad (1)$$

where W_{reg} contains the network weights and dimensionless energy-readout weights, excluding biases and radial parameters. We use energy-only Adam training, learning rate 10^{-3} , $(\beta_1, \beta_2) = (0.9, 0.999)$, and batches of 30. The rate remains unchanged for at least 100 epochs and halves after 50 epochs without a better validation MAE; training stops after 100 unimproved epochs or at 2,000 epochs. The checkpoint with the lowest validation MAE is retained. No test statistic controls this schedule. Single-precision training disables TF32. A dense environment contraction and CUDA graph replay accelerate the original hippynn modules; comparison with its sparse implementation verifies energies, R , and parameter gradients. Epoch logs include online minibatch training MAE and end-of-epoch validation MAE. Three seeds quantify variation from initialization and the validation partition, rather than the eight random splits used in the original paper.

2.2 What a squared ratio cannot identify

For an atom i , denote its learned contributions by e_i^0, e_i^1, e_i^2 , and define

$$q_{in} = \frac{(e_i^n)^2}{(e_i^{n-1})^2 + (e_i^n)^2}, \quad R = \sum_i (q_{i1} + q_{i2}). \quad (2)$$

These are the original atomic ratios, not ratios formed after molecular summation. Interaction order is a receptive-field hierarchy, not an exact body-order expansion. The decomposition is learned and nonunique.

Sign and scale ambiguity. Keeping all $|e_i^n|$ fixed while changing signs leaves R unchanged but can change the molecular energy. Multiplying every contribution by a common $a \neq 0$ also leaves R unchanged, whereas the energy scale changes by $|a|$. Both statements follow immediately from the squares in Eq. (2). Thus R alone does not algebraically determine either energetic scale or cancellation. This is an identifiability observation about a diagnostic, not a constructed pair of physically realizable molecular counterexamples.

To retain an energy scale without the pole of a geometric-series extrapolation, define a damped continuation magnitude

$$t_i = |e_i^2| \sqrt{q_{i2}}, \quad L = \sum_i t_i, \quad T = \max_i t_i, \quad C = 1 - \frac{|\sum_i e_i^2|}{\sum_i |e_i^2| + \epsilon}, \quad (3)$$

with $\epsilon = 10^{-12} \text{ kcal mol}^{-1}$. When $|e_i^2| \ll |e_i^1|$, t_i approximates $(e_i^2)^2/|e_i^1|$, the next magnitude of a locally geometric decay. When decay fails, $t_i \leq |e_i^2|$ remains finite. L measures aggregate continuation demand, T its strongest atomic concentration, and LC the part associated with opposing final-order contributions. No omitted energy order or error bound is assumed to exist.

We call the resulting diagnostic the *Continuation and Cancellation Audit* (CCA). On the fit partition, normalize each component of $x = (R, L, LC, T)$ by its median (floored at 10^{-12}), then fit

$$s(x) = \beta_0 + \sum_{j=1}^4 \beta_j x_j / m_j, \quad \hat{\beta} = \arg \min_{\beta \geq 0} \frac{1}{N_f} \sum_{k \in \mathcal{F}} (s(x_k) - |\hat{E}_k - E_k|)^2 + 10^{-3} \sum_{j=1}^4 \beta_j^2. \quad (4)$$

The intercept is unpenalized. A nonnegative coefficient can become zero, providing a directly inspectable ablation of that signal. On the disjoint calibration partition, isotonic least-squares regression maps s to $u = g(s)$; values outside its score range receive the corresponding endpoint prediction. The energy network is frozen throughout. Comparators are a constant calibration-set mean error, isotonic R , isotonic L , and the uncalibrated linear score. Each comparator uses the same calibration molecules. No uncertainty head, ensemble, dropout realization, or extra energy forward pass enters CCA.

2.3 Motifs, multiplicity, and compactness

The twelve candidates in Table 2 span ring constraints, bond-order organization, heteroatom arrangements, flexibility, and spatial proximity. Definitions operate on the supplied relaxed SMILES using RDKit [RDKit contributors, 2026]; only close N/O contacts additionally require coordinates. Rings use the symmetrized smallest-ring set, rotatable bonds use RDKit’s strict definition, and conjugation uses its aromaticity/bond flags. A close contact joins N/O atoms less than 2.8 Å apart but at least three graph bonds apart; its graph connects pairs within 1.25 times the sum of RDKit covalent radii, excluding separations below 0.4 Å. Definitions use neither energy errors nor hierarchy outputs.

Each motif receives separate one-sided Mann–Whitney tests for upward rank shifts in absolute error and R relative to nonmatching molecules. We average these quantities within canonical connectivity graphs before testing, preventing repeated graphs from acting as independent units. If p_E and p_R are the two p -values, $p_{\text{joint}} = \max(p_E, p_R)$ tests the joint claim that *both* are elevated. Holm correction [Holm, 1979] controls family-wise error at 0.05 across all twelve motifs and all three seeds: 36 hypotheses, including null or unfavorable candidates. No evaluation-derived motif is added. Per-molecule coverage, median, upper error quantiles, and seed-specific statistics are retained independently of significance.

The primary failure library includes only candidates whose *fit-partition* mean error and mean R both exceed their nonmatching means. This direction-based discovery step makes no significance claim. Candidates are ranked by the number of fit-partition errors above 1 kcal mol^{−1} that they match. On final evaluation, cumulative attribution counts the *union* of matching failures, with *all* failures in the denominator. K_{80} is the smallest rank reaching 80%; it is undefined if the entire selected library falls short. We also report molecule prevalence for the same unions and all-candidate coverage. This distinguishes a compact error explanation from a broad description of most molecules. Atom-localization summaries compare q_{i2} on motif atoms with other atoms in the same molecule; graph-to-coordinate mapping succeeds for 130,301 molecules. These summaries are mechanistic associations, not causal interventions.

Exploratory sensitivity checks additionally tabulate disjoint ring-motif intersections and standardize matching/nonmatching means to identical molecular-formula frequencies. Formula strata must contain at least five molecules in each group; formulas with no structural contrast are not extrapolated. These checks were added after preliminary descriptive results and do not select candidates, tune CCA, or supply additional significance claims. Rank-effect sizes and order truncations are likewise descriptive.

2.4 Calibration and conformational transfer

We sort final-evaluation molecules by predicted u , divide them into ten equal-count bins, and report every bin, including both tails. Equal-score ties use a fixed independent random ordering (seed 20261004). This avoids accidentally stratifying tied predictions by QM9’s chemically ordered IDs; the tie rule was corrected during implementation review, without optimizing against calibration outcomes. The plotted abscissa is the bin’s actual mean prediction. With $a_k = |\hat{E}_k - E_k|$,

$$\text{ECE}_{\text{abs}} = \sum_{b=1}^{10} \frac{N_b}{N} |\bar{u}_b - \bar{a}_b|. \quad (5)$$

This ECE has units of kcal/mol. We additionally report $\text{ECE}_{\text{abs}}/\text{MAE}$, ROC-AUC, average precision, and retained-error curves. A constant predictor is included because a small ECE alone does not establish useful discrimination. High error always means $a > 1$ kcal mol^{−1}; the fixed operating point flags $u > 1$ kcal mol^{−1}. Precision, recall, false-positive and false-negative rates, and a full threshold sweep are computed on the final partition. This operating point aligns an expected-error scale with the application threshold; it is not chosen to maximize evaluation F1. A prespecified descriptive label-budget analysis uses nested random subsets of 250, 1,000, or 5,000 molecules from each of the fit and calibration partitions, preserving the same final evaluation set. Unless explicitly identified as a count, a p -value, or an individual example, results are means $\pm$ standard errors over the three energy-model seeds. These bars describe between-fit variation, not population confidence intervals.

Table 1: Published HIP-NN anchors and the unchanged energy model audited here. N includes optimizer-training and validation examples. Published MAEs use random full-QM9 tests; ID MAE here uses the independent smaller-molecule audit. Nine-heavy MAE uses all 109,024 molecules; ECE and K_{80} use only final evaluation. NR means that 80% coverage is not reached. Errors are kcal/mol, with mean $\pm$ seed standard error. Published values are Table I of [Lubbers et al. \[2018\]](#).

Method	N	Random/ID MAE	Nine-heavy MAE	ECE	K_{80}
HIP-NN (reported)	110,426	0.256 ± 0.003	—	—	—
HIP-NN (reported)	50,000	0.354 ± 0.004	—	—	—
HIP-NN (reimplemented)	19,799	0.317 ± 0.007	0.780 ± 0.004	—	—
Same HIP-NN + CCA	19,799	0.317 ± 0.007	0.780 ± 0.004	0.030 ± 0.004	1,1,1

ANI-1x [\[Smith et al., 2020\]](#) supplies a separate conformational test. We uniformly sample 60,000 configurations without replacement from the 4,956,005 finite `wb97x_dz.energy` entries, using seed 31415. The HDF5 sampling follows the official loader’s finite-energy convention [\[ANI1x dataset contributors, 2020\]](#). Graphs are inferred by RDKit’s coordinate-based bond assignment, with van der Waals connectivity and no Hückel calculation. Disconnected or unassignable graphs are recorded and excluded from graph-based analysis. Canonical connectivities represented by at least six sampled conformers are retained; a seeded random half estimates one molecular offset and the remainder provides evaluation. Thus the transfer population is the eligible subset of a representative initial sample, not all ANI conformations.

Because ANI uses a different functional and basis from QM9, we evaluate

$$a_{mc}^{\text{rel}} = \left| \hat{E}_{mc} - E_{mc}^{\text{ANI}} - \text{median}_{c' \in \mathcal{O}_m} (\hat{E}_{mc'} - E_{mc'}^{\text{ANI}}) \right|, \quad (6)$$

where $\mathcal{O}_m$ is the disjoint offset-fitting half for connectivity m . This removes a constant functional/composition offset, including the atomization convention, but not geometry-dependent functional differences. QM9 U_0 also includes a zero-point contribution, whereas the ANI target is electronic energy; constant components of this difference are removed, but the transfer test cannot disentangle every target-definition effect. It measures relative-energy *discrepancy*, not pure same-functional model error. Motif transfer rates are the fractions with $a^{\text{rel}} > 1 \text{ kcal mol}^{-1}$, compared with nonmatching conformers. Inference uses one mean discrepancy per graph, with a motif present when at least half of that graph’s evaluation conformers match. One-sided error tests undergo a separate Holm correction across twelve motifs and three seeds; unstable motifs receive $p = 1$. Size-stratified effects distinguish transfer from a shift in molecule size. No ANI energy updates HIP-NN or the QM9 calibrator. All energy prediction remains force-free, and no new quantum calculations are performed.

3 Chemical structure in the extrapolation error

3.1 A strong smaller-molecule baseline leaves a substantial tail

Table 1 separates the published random-split anchors from our independent audit. ID MAEs are 0.3081, 0.3298, and 0.3121 kcal/mol for seeds 11, 22, and 33, respectively, yielding 0.317 ± 0.007 . Each improves on the published 50,000-example result despite using fewer training examples. This is a useful reproduction check, but the smaller-molecule test is easier and does not establish a better full-QM9 learning curve. In particular, it does not improve on the published 110,426-example MAE of 0.256 ± 0.003 kcal/mol. A power law through the two published anchors, $A(N) = 0.354(50,000/N)^{0.409}$, gives a descriptive size-adjusted reference of 0.517 kcal/mol at our combined training/validation size. This extrapolation is not an experiment on a matched chemical distribution.

All runs terminate through validation patience at epochs 1574, 1226, 1811, retaining checkpoints 1474, 1126, 1711. They use the full architecture and scheduled convergence criterion, rather than a shortened training budget. Table 6 and Fig. 7 give the complete convergence evidence. Across all nine-heavy-atom molecules, MAE rises to 0.780 ± 0.004 kcal/mol. On final evaluation, $18.1 \pm 0.3\%$ of errors exceed 1 kcal/mol. The original paper’s $2.26 \pm 0.07\%$ failure rate concerns its random full-QM9 split [\[Lubbers et al., 2018, Table II\]](#); it is not a size-extrapolation control. The error- R Spearman correlation here is 0.343 ± 0.001 , establishing a useful but incomplete ordering of risk.

Table 2: The complete candidate library, including null and unfavorable associations. Coverage is on final-evaluation molecules; MAE and P_{90} describe matching molecules (kcal/mol). R ratio compares matching with nonmatching means. Support gives significant seeds/3 and the largest Holm-adjusted joint rank-test p across seeds; rank shifts need not imply larger means. Definitions are outcome-independent; details are in Section 2.

Motif and definition	N (%)	MAE	P_{90}	R ratio	Support; $p_{\max}$
Small ring 3- or 4-membered ring	27,381 (62.8)	0.84 ± 0.01	1.55 ± 0.03	1.58 ± 0.05	3/3; 3.1×10^{-16}
Polycyclic ≥ 2 SSSR rings	23,175 (53.1)	1.00 ± 0.01	1.90 ± 0.03	1.93 ± 0.05	3/3; $< 10^{-99}$
Unsaturated ring Aromatic/multiple ring bond	13,650 (31.3)	0.87 ± 0.01	1.80 ± 0.02	0.91 ± 0.04	3/3; 5.6×10^{-7}
Extended conjugation ≥ 3 conjugated bonds	11,861 (27.2)	0.808 ± 0.004	1.69 ± 0.01	0.82 ± 0.06	2/3; 0.235
Multiple carbonyls ≥ 2 C=O bonds	2,140 (4.9)	0.67 ± 0.01	1.36 ± 0.01	0.70 ± 0.01	0/3; 1.000
Heteroatom adjacency Direct N/O–N/O bond	4,085 (9.4)	0.98 ± 0.01	2.06 ± 0.02	1.08 ± 0.15	3/3; 1.0×10^{-46}
Heteroatom crowding Heavy center with ≥ 2 N/O neighbors	16,425 (37.7)	0.872 ± 0.004	1.71 ± 0.01	0.96 ± 0.05	3/3; 1.5×10^{-69}
Fluorinated At least one F atom	665 (1.5)	0.67 ± 0.01	1.48 ± 0.02	0.80 ± 0.05	0/3; 1.000
Flexible ≥ 3 strict rotatable bonds	4,449 (10.2)	0.37 ± 0.01	0.73 ± 0.02	0.52 ± 0.01	0/3; 1.000
Close heteroatoms N/O distance < 2.8 Å; graph distance ≥ 3	5,555 (12.7)	0.829 ± 0.003	1.58 ± 0.04	0.97 ± 0.02	3/3; 4.1×10^{-7}
Heterocycle Ring contains N or O	31,280 (71.7)	0.89 ± 0.01	1.69 ± 0.02	1.42 ± 0.04	3/3; $< 10^{-99}$
Triple bond At least one triple bond	10,762 (24.7)	0.477 ± 0.005	0.99 ± 0.01	0.71 ± 0.03	0/3; 1.000

3.2 Seven reproducible rank associations, three dominant mean-risk motifs

Table 2 reports all twelve candidates. Small rings, polycycles, unsaturated rings, N/O adjacency, N/O crowding, close N/O contacts, and heterocycles pass the joint error- R rank test in every seed. Extended conjugation passes in two seeds; multiple carbonyls, fluorination, flexibility, and triple bonds pass in none. Coverage ranges from 665 fluorinated molecules to 31,280 heterocycles, so absence of support is not concealed by dropping uncommon or unfavorable candidates.

The distinction between rank shifts and arithmetic means matters. Small rings, polycycles, and heterocycles have mean-error ratios of 1.26 ± 0.01 , 1.88 ± 0.01 , and 1.74 ± 0.02 , respectively, relative to nonmatching molecules. They also show larger mean R in every seed. Unsaturated rings, adjacency, crowding, and close contacts can have significantly larger typical ranks even when their *mean* R is lower. Rank tests and arithmetic means weight distribution tails differently; an upward rank association need not shift the entire distribution. The probabilities underlying the rank tests and the median ratios are reported in Table 12. Statistical support therefore does not justify describing every motif as a source of the largest hierarchy values.

Where the hierarchy carries the burden. Figure 1 separates energetic association, total non-hierarchicality, and atomic localization. For small rings, the matching/nonmatching ratio of $Q_2 = \sum_i q_{i2}$ is 2.12 ± 0.14 , compared with 1.12 ± 0.06 for Q_1 . Polycycles amplify this difference: 2.83 ± 0.12 versus 1.22 ± 0.04 . A plausible structural mechanism is that constrained distances and multiple paths through the same local environment demand a substantial final interaction correction. The observation supporting that interpretation is the enlarged second-order ratio and its concentration on ring atoms, rather than ring strain being independently measured. The model receives distances and elements, not explicit bond or ring labels.

Heterocycles combine ring constraints with chemically different neighboring atoms. Their Q_1 and Q_2 ratios are 1.34 ± 0.07 and 1.49 ± 0.10 , indicating burden at both interaction stages. In an exploratory check at identical molecular-formula frequencies, the small-ring, polycyclic, and heterocycle error ratios are 1.49 ± 0.01 , 2.11 ± 0.02 , and 1.74 ± 0.02 (Table 13). Their standardized R ratios are 1.72 ± 0.03 , 1.97 ± 0.04 , and 1.54 ± 0.03 . These restricted comparisons reduce composition confounding but cannot isolate a causal ring effect from correlated substructures.

Unsaturated rings provide a different signature. Their error ratio is 1.19 ± 0.01 , while the average Q_2 ratio is 0.83 ± 0.06 ; positive atom-local contrasts coexist with a less extreme molecular tail. Bond-order organization must be inferred from geometry, so residual difficulty with coupled environments is plausible, but these data do not establish a general failure on conjugated electrons. Indeed, extended conjugation is less stable across seeds, and multiple carbonyls are not an elevated-error class under the confirmatory rule.

Formula standardization reverses the unsaturated-ring error association to 0.87 ± 0.02 . N/O adjacency likewise changes to 0.92 ± 0.01 , with standardized R ratio 1.00 ± 0.08 . These reversals restrict the interpretation of the unadjusted markers: their occurrence predicts risk in this evaluation population, but the corresponding bond pattern is not established as the independent cause. In contrast, the standardized close-contact error and R ratios remain above one, at 1.235 ± 0.002 and 1.28 ± 0.01 .

N/O adjacency raises the first-order ratio by a factor of 1.28 ± 0.08 , consistent with an increased correction to the element-only baseline when neighboring polar environments interact. Crowding shows a smaller first-order increase, 1.05 ± 0.01 . Their second-order behavior varies across seeds, emphasizing that a learned decomposition can distribute corrections differently while retaining similar energy error. Close N/O contacts, defined spatially across at least three graph bonds, identify interactions between otherwise separated graph neighborhoods. Their mean-error ratio is only 1.07 ± 0.01 , and their Q_2 mean ratio is 0.93 ± 0.04 ; localized second-order excess and rank associations support a modest environmental diagnostic, not a universal high- R class or a hydrogen-bond assignment.

These interpretations remain associations. Localization compares motif atoms with all other atoms, including hydrogens, and the energy decomposition has no unique physical gauge. Removing higher orders without refitting increases evaluation MAE from 0.780 ± 0.005 to 3.02 ± 0.09 kcal/mol when order two is removed and to 20.404 ± 0.005 for the fixed atomic baseline alone. Higher orders are necessary for accuracy; their necessity does not make each atomic correction a physical interaction energy.

3.3 A compact vocabulary has limited specificity

The discovery rule selects 3, 3, 5 motifs for the three seeds. Heterocycles, polycycles, and small rings are selected in the same order in all three; seed 33 additionally selects crowding and adjacency. Heterocycles alone reach $85.4 \pm 0.2\%$ failure attribution, making $K_{80} = 1$ in every run (Fig. 2). They also occur in 71.7% of evaluation molecules, a failure-coverage lift of only 1.191 ± 0.002 . The first three motifs cover $95.7 \pm 0.1\%$ of failures but 87.7% of molecules, reducing the lift to 1.090 ± 0.001 . The full seed-specific libraries cover $96.5 \pm 1.0\%$ of failures and $89.2 \pm 1.5\%$ of molecules.

These unions do not double-count molecules, and uncovered failures remain in the denominator. The disjoint intersections in Table 14 show that the ring motifs are overlapping descriptions, not independent mechanisms. Molecules matching only the small-ring candidate among these three have MAE 0.360 ± 0.004 kcal/mol, compared with 0.419 ± 0.003 when none is present. Thus the marginal small-ring association does not imply uniformly increased risk outside its co-occurring motifs. Compactness here means that a short chemical vocabulary summarizes much of the tail. It offers weak selectivity for deciding which individual predictions to trust. The gray all-candidate curve is a prevalence control; its further coverage is not evidence that null motifs explain failure.

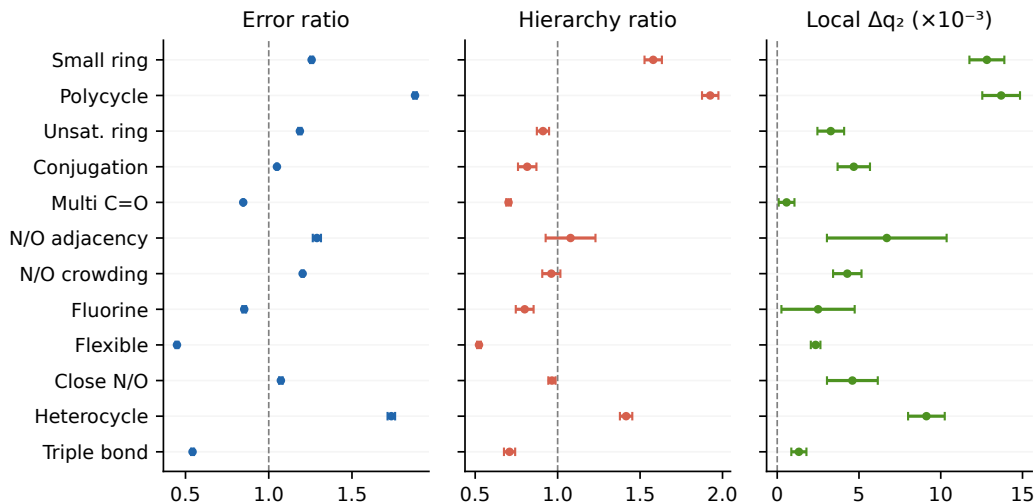


Figure 1: Chemical association and order localization on final QM9 evaluation. Ratios compare matching and nonmatching means; the local contrast compares q_{i2} on motif atoms with other atoms in the same molecule. Points and bars are seed means and standard errors. Mean ratios and the corrected rank tests answer different questions, explaining supported motifs with ratios below one.

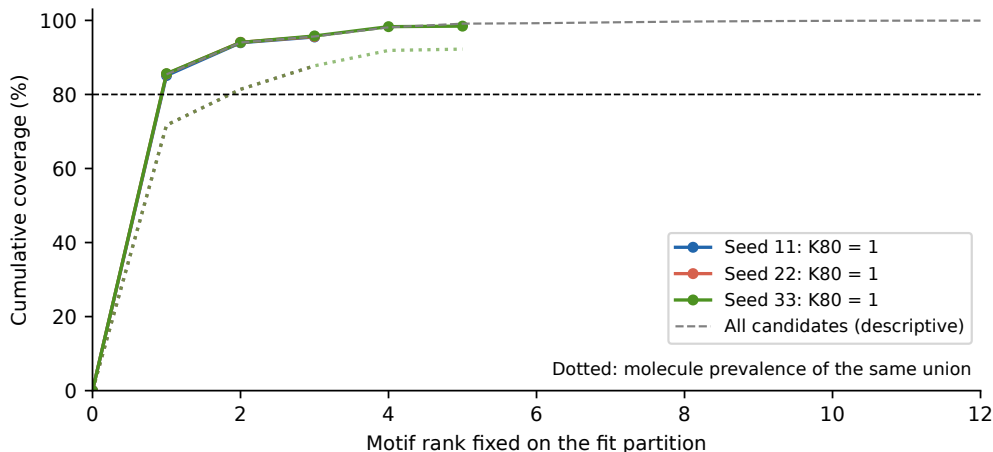


Figure 2: Failure attribution by discovery-ranked motifs. Solid curves count the union of all evaluation failures covered; dotted curves show the prevalence of the identical unions. Each colored curve ends at that seed’s selected library size. The gray curve includes every candidate and is descriptive. Reaching 80% at one motif reflects both enrichment and the high prevalence of heterocycles.

4 Calibrated means remain incomplete warnings

4.1 Continuation adds discrimination, not uniformly better calibration

CCA attains ECE 0.030 ± 0.004 kcal/mol, or normalized ECE 0.039 ± 0.005 , on held-out chemistry (Table 3). Every one of the ten bin means is within 10% of observed error in each seed; Fig. 3 includes both tails, and Table 8 gives all seed-resolved values. This is a statement about bin means, not every uncertainty value or molecule. Isotonic R is similarly calibrated, with ECE 0.030 ± 0.006 . CCA increases ROC-AUC from 0.746 ± 0.005 to 0.757 ± 0.004 and average precision from 0.445 ± 0.004 to 0.461 ± 0.002 . These modest differences are descriptive across three fits, without an additional method-comparison significance claim.

Table 3: Held-out uncertainty quality. ECE is in kcal/mol; nECE divides by evaluation MAE. AP is average precision for errors above 1 kcal/mol. All rows use the same frozen energy predictions and final evaluation molecules.

Diagnostic	ECE	nECE	ROC-AUC	AP
Constant mean	0.027 ± 0.005	0.035 ± 0.006	0.500 ± 0.000	0.181 ± 0.003
Isotonic R	0.030 ± 0.006	0.039 ± 0.007	0.746 ± 0.005	0.445 ± 0.004
Isotonic L	0.028 ± 0.009	0.035 ± 0.011	0.753 ± 0.005	0.459 ± 0.005
CCA before calibration	0.036 ± 0.016	0.047 ± 0.020	0.758 ± 0.004	0.474 ± 0.002
CCA	0.030 ± 0.004	0.039 ± 0.005	0.757 ± 0.004	0.461 ± 0.002

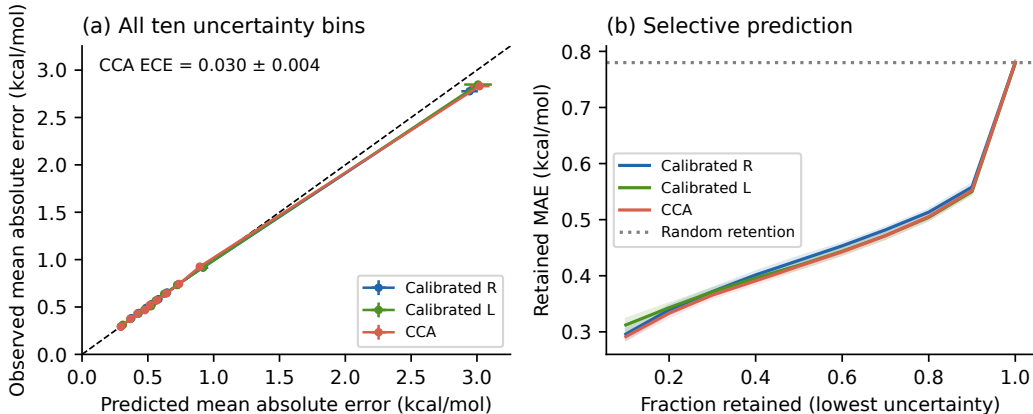


Figure 3: Calibration and selective prediction. Left: all ten equal-count bins, averaged by bin rank across seeds, with standard errors on both axes and the identity line. Right: error after retaining low-uncertainty molecules, with seed-standard-error bands. Individual-seed bin means are reported in Table 8. Global calibration does not imply reliable individual warnings.

The ablations constrain the interpretation of the extension. In every fit, the nonnegative regression retains R and LC , while the separate coefficients on L and T are exactly zero (Table 7). Thus aggregate scale without cancellation and the maximum local term add no linear contribution under this fit criterion. Isotonic L alone remains a competitive diagnostic, and CCA is not uniformly best across ECE, precision, recall, or seeds. Calibration also merges score ranks into tied plateaus, reducing average precision relative to the uncalibrated linear score even as it adjusts the error scale.

The constant-mean control has low ECE but ROC-AUC 0.5 and no warnings at the chosen threshold. Consequently, a calibration target alone would reward a diagnostic with no ability to distinguish molecules. CCA instead reduces MAE to 0.417 ± 0.006 kcal/mol when retaining the half of evaluation molecules with lowest uncertainty (Fig. 3b). Its usefulness is selective ranking with a calibrated average scale, rather than guaranteed detection of the tail.

4.2 Missed failures persist at a justified operating point

At $u > 1$ kcal/mol, CCA flags $11.2 \pm 1.2\%$ of molecules with precision $54.4 \pm 2.6\%$ and recall $33.2 \pm 2.0\%$ (Table 4). Its false-positive rate is $6.3 \pm 1.0\%$, and its false-negative rate is $66.8 \pm 2.0\%$. This threshold asks whether predicted *mean* error exceeds the application tolerance; it does not optimize a classifier for the event $a > 1$. A distribution with a low mean can retain a substantial high-error tail. The full threshold sweep in Fig. 4 makes the resulting precision–recall tradeoff explicit.

Calibration also has a label cost. Restricting fit and calibration data to 250 molecules each raises ECE to 0.114 ± 0.004 kcal/mol; 1,000 each yields 0.074 ± 0.018 , and 5,000 each yields 0.039 ± 0.001 (Table 9). The main CCA model uses all 43,609 fit and 21,799 calibration labels. The energy predictor extrapolates without nine-heavy-atom training energies, but the uncertainty scale is adapted using labeled nine-heavy-atom chemistry; this is not zero-label uncertainty transfer.

Table 4: The fixed operating point $u > 1$ kcal/mol, evaluated against actual error > 1 kcal/mol. All entries are percentages, mean $\pm$ seed standard error. FPR and FNR condition on the true negative and positive classes, respectively.

Diagnostic	Flagged	Precision	Recall	FPR	FNR
Isotonic R	12.7 ± 1.2	50.4 ± 1.9	35.1 ± 2.0	7.7 ± 1.0	64.9 ± 2.0
Isotonic L	12.0 ± 1.4	53.4 ± 2.4	35.0 ± 2.0	6.9 ± 1.1	65.0 ± 2.0
CCA	11.2 ± 1.2	54.4 ± 2.6	33.2 ± 2.0	6.3 ± 1.0	66.8 ± 2.0

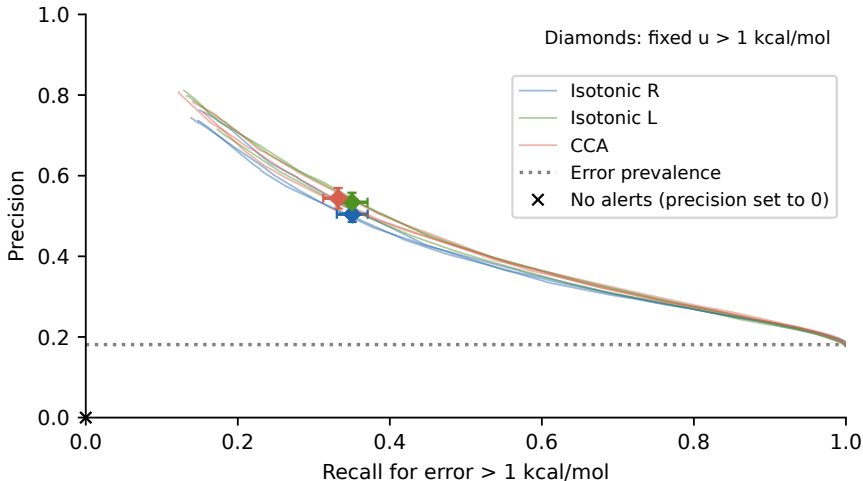


Figure 4: Threshold sweeps for the same error event, $a > 1$ kcal/mol. Thin curves show the three individual seeds; diamonds summarize the fixed $u > 1$ kcal/mol operating point. Thresholds producing no alerts have precision set to zero and are shown separately at the origin, without interpolating to nonempty alert sets. The horizontal reference is observed failure prevalence.

5 Conformational transfer preserves only part of the library

The representative initial ANI sample yields 49,735 connected parsable conformers and 10,265 excluded cases. Another 31,942 belong to connectivities sampled too sparsely for the offset split. The resulting panel contains 1,719 graphs, 8,549 offset-fitting conformers, and 9,244 evaluation conformers. This eligibility filter favors repeatedly sampled connected molecules; inference is conditional on that panel. The HDF5 composition buckets are not treated as molecular identities.

Mean relative-energy discrepancy is 19.90 ± 0.15 kcal/mol, with $95.9 \pm 0.2\%$ above 1 kcal/mol. A descriptive constant-energy-per-connectivity control, centered using the same offset half, has discrepancy 22.07 kcal/mol. The improvement over this constant baseline is modest, underscoring the limited conformational accuracy despite a strong QM9 equilibrium result. These discrepancies mix potential error with geometry-dependent reference-method differences; subtracting a molecular offset does not make the two electronic-structure methods identical.

Small rings, polycycles, heterocycles, and close N/O contacts retain significant positive associations in all three seeds (Table 5, Fig. 5). Adjacency transfers in two seeds, crowding and unsaturated rings in one, while conjugation, multiple carbonyls, flexibility, and triple bonds show no consistent elevated-discrepancy support. Fluorination is absent and untestable. Because most conformers already exceed 1 kcal/mol, per-motif binary rates are close to saturation; the graph-level discrepancy contrasts are more informative about transfer strength.

Topology-based definitions remain applicable along conformations of a fixed graph, but their energy association need not persist. The close-contact motif is explicitly conformation dependent: 566 eligible graphs change contact status across their evaluation conformers. Its overall association survives, yet size stratification reveals limits (Table 16). In particular, the aggregate positive close-contact effect does not hold uniformly in the nine-heavy or larger strata. The panel is predominantly

Table 5: ANI-1x transfer on 1,719 eligible connectivities. N_m counts majority-matching graphs. Discrepancies are means of graph means (kcal/mol); rates are percentages of matching/nonmatching evaluation conformers exceeding 1 kcal/mol. The final column gives significant seeds/3 after the separate 36-test Holm correction. Fluorination is absent from HCNO chemistry and is untestable.

Motif	N_m	Discrepancy $\pm$		Rate $\pm$ (%)		$p_{\max}$	Support
Small ring	319	25.91 $\pm$ 0.43	20.35 $\pm$ 0.14	96.7 $\pm$ 0.3	95.7 $\pm$ 0.2	6.5×10^{-8}	3/3
Polycyclic	253	29.68 $\pm$ 0.39	19.95 $\pm$ 0.19	97.2 $\pm$ 0.2	95.7 $\pm$ 0.2	1.0×10^{-23}	3/3
Unsaturated ring	788	21.42 $\pm$ 0.20	21.35 $\pm$ 0.23	96.4 $\pm$ 0.1	95.5 $\pm$ 0.2	1.000	1/3
Extended conjugation	1141	20.89 $\pm$ 0.17	22.37 $\pm$ 0.34	96.0 $\pm$ 0.1	95.7 $\pm$ 0.4	1.000	0/3
Multiple carbonyls	218	19.39 $\pm$ 0.45	21.67 $\pm$ 0.23	96.2 $\pm$ 0.1	95.9 $\pm$ 0.2	1.000	0/3
Heteroatom adjacency	783	21.80 $\pm$ 0.28	21.04 $\pm$ 0.28	96.5 $\pm$ 0.1	95.4 $\pm$ 0.2	0.423	2/3
Heteroatom crowding	1109	21.36 $\pm$ 0.18	21.43 $\pm$ 0.33	96.3 $\pm$ 0.1	95.2 $\pm$ 0.3	0.628	1/3
Fluorinated	0	not testable		not testable		1	0/3
Flexible	222	21.81 $\pm$ 0.65	21.32 $\pm$ 0.25	96.1 $\pm$ 0.3	95.9 $\pm$ 0.1	1.000	0/3
Close heteroatoms	565	23.26 $\pm$ 0.11	20.47 $\pm$ 0.23	96.8 $\pm$ 0.2	95.5 $\pm$ 0.2	1.5×10^{-5}	3/3
Heterocycle	966	22.64 $\pm$ 0.20	19.77 $\pm$ 0.17	96.5 $\pm$ 0.1	95.3 $\pm$ 0.2	5.7×10^{-6}	3/3
Triple bond	153	18.48 $\pm$ 0.27	21.67 $\pm$ 0.18	93.9 $\pm$ 0.7	96.1 $\pm$ 0.2	1.000	0/3

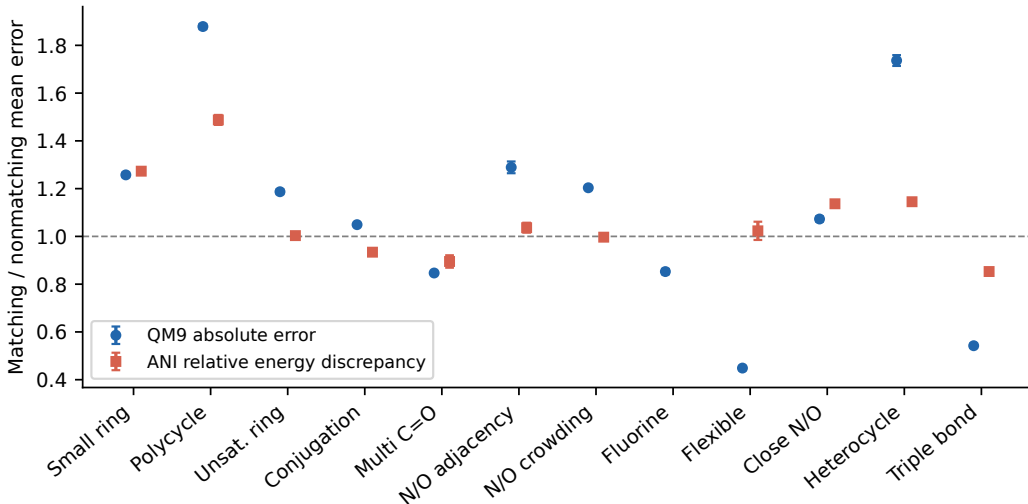


Figure 5: Conditional effect sizes on optimized QM9 and off-equilibrium ANI geometries. QM9 ratios use per-molecule means; ANI ratios use means of connectivity-level mean discrepancies. Both show matching/nonmatching contrasts with seed standard errors, but the targets and populations differ. No ANI value is plotted for fluorination. Corrected transfer tests and all rates appear in Table 5.

within the model’s size range: 86.9% of evaluation conformers have at most eight heavy atoms. There are only 27 nine-heavy-atom graphs and 175 corresponding evaluation conformers, including no polycyclic graph in that stratum. Thus this is substantial evidence about off-equilibrium HCNO transfer, but weak evidence for an isolated nine-heavy-atom conformational effect.

The error- R Spearman correlation falls to 0.185 ± 0.017 , compared with 0.343 ± 0.001 on QM9 evaluation. Ring constraints and spatially close heteroatoms retain an interpretable association, while the overall scale of error and the weaker hierarchy correlation expose the limits of an equilibrium-trained energy hierarchy. No force prediction or trajectory-stability claim follows from these measurements.

6 What the diagnostic still misses

The selected libraries leave $3.5 \pm 1.0\%$ of evaluation failures uncovered. In seeds 11 and 22, the 5,351 uncovered molecules have MAEs of 0.416 and 0.418 kcal/mol, respectively, yet include severe exceptions. Figure 6 illustrates distinct failure mechanisms of the audit, with categories defined using

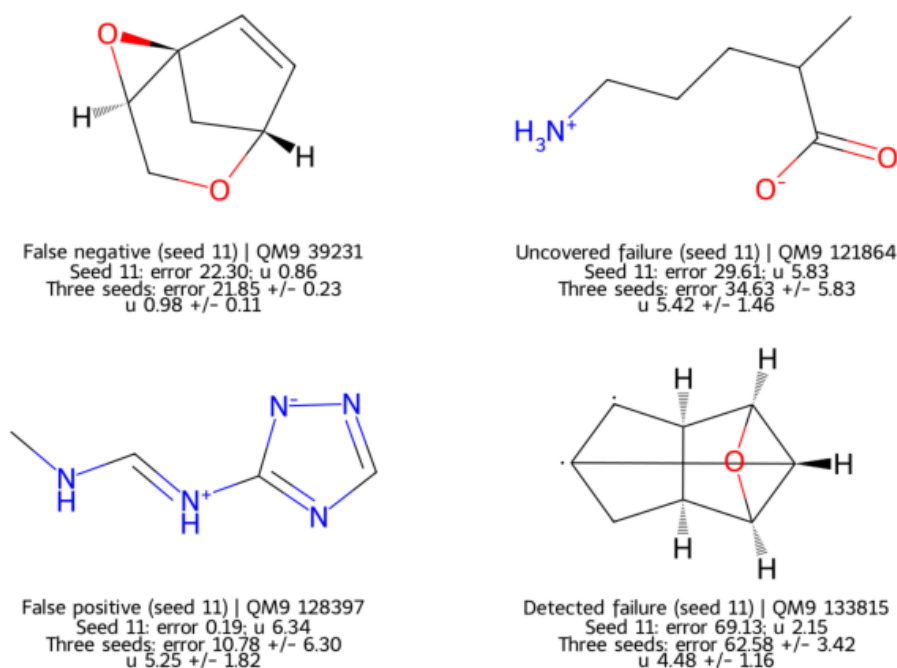


Figure 6: Descriptive failure cases drawn from the supplied QM9 molecular graphs. Category labels use seed 11; each panel states that seed’s error and uncertainty and their means $\pm$ standard errors across all three fits (kcal/mol). The apparent false positive is highly model dependent, and the uncovered case is flagged by CCA. These selected illustrations are not estimates of prevalence.

seed 11 and examples chosen descriptively by largest mean error across fits. These examples never redefine a motif or enter a new significance test.

QM9 molecule 121864 is an acyclic, charge-separated structure in the supplied optimized SMILES. Its error is 34.63 ± 5.83 kcal/mol. It matches crowding and flexibility but none of the three ring motifs selected by seeds 11 and 22. CCA nevertheless flags it, with mean uncertainty 5.42 ± 1.46 kcal/mol. It becomes covered when seed 33’s library includes crowding. This example distinguishes missing chemical vocabulary from missing an energy warning, and suggests formal-charge organization as a hypothesis for a separately designed future study.

The converse occurs for molecule 39231, which matches small rings, polycycles, unsaturated rings, and heterocycles. Its error is 21.85 ± 0.23 kcal/mol, but uncertainty is 0.98 ± 0.11 kcal/mol; it is a false negative in seeds 11 and 33. A recognizable failure-associated substructure does not force a large predicted error for every instance. Molecule 133815 is detected in all fits, but its error, 62.58 ± 3.42 kcal/mol, far exceeds its mean uncertainty of 4.48 ± 1.16 kcal/mol. This is compatible with mean calibration and demonstrates why u cannot be interpreted as an upper bound.

False alarms are also real and model dependent. Molecule 128397 receives $u = 6.34$ kcal/mol with actual error 0.19 kcal/mol in seed 11, while its errors in the other fits are 10.17 and 21.97 kcal/mol. Across the complete evaluation set, pairwise warning-set Jaccard similarities range from 0.395 to 0.424, despite agreement on the three dominant ring motifs. Thus the chemical summary is more stable than the exact list of flagged molecules. Neither the small seed variation in global ECE nor a statistically supported motif resolves this molecule-level instability.

6.1 Cost and scope of the evidence

Each energy fit takes 59.7 ± 4.0 minutes of wall time, with the three jobs initially sharing one NVIDIA H200. The actual hardware differs from an H100; these timings should not be read as

an H100 performance claim. Isolated resident-input inference at batch size 128, on the first 1,024 final-evaluation molecules, costs 6.30 ± 0.02 microseconds per molecule for the energy and hierarchy. CPU continuation, cancellation, and isotonic postprocessing cost 1.13 ± 0.34 microseconds per molecule. Parsing a supplied SMILES and assigning all twelve motifs costs 0.520 ms per molecule on CPU. These measurements exclude model loading and dataset decompression, and describe batched throughput rather than single-request latency. CCA adds no energy-network pass, but its labeled calibration data and chemistry parsing are substantive costs. Structure-only assignments can be cached for repeated conformations; the distance-based contact still requires recomputation.

Several limitations bound the claims. The energy benchmark uses three fits and an extra independent ID holdout, and its smaller test distribution differs from the original random full-QM9 benchmark. The disclosed interpretation of the paper’s initialization scale is another implementation choice. All fits share a fixed chemistry evaluation partition, so seed standard errors do not quantify uncertainty from drawing a new chemical population. The fixed motif family is finite and overlapping, and significant rank effects can be small or composition dependent. Localization and the continuation formula characterize a learned decomposition rather than identify unique physical energy terms.

The calibration experiment is adaptation within labeled nine-heavy-atom chemistry; exchangeability with MD, larger molecules, or a different quantum target is not assumed. ANI provides an off-equilibrium challenge, but graph reconstruction, repeated-conformer eligibility, and functional mismatch restrict its interpretation. There is no fluorine transfer test, no same-functional conformational reference, no force supervision, and no trajectory validation. The large ANI discrepancy cannot be partitioned into model and reference-method contributions with the available labels. The retained source configurations and exclusion reasons make these limitations auditable.

6.2 Conclusion

HIP-NN’s hierarchy reveals reproducible chemical associations under size extrapolation. Ring constraints produce the clearest persistent mean-error and second-order signatures; other statistically supported markers are weaker, composition dependent, or less portable. A short vocabulary covers most failures largely because its motifs are common. Continuation and cancellation improve discrimination modestly over the original ratio and permit accurate held-out bin-mean calibration, yet many individual high-error predictions remain unflagged. Off-equilibrium transfer preserves four motif associations while substantially degrading energy accuracy. The hierarchy is therefore useful as a chemical audit and a calibrated ranking signal, with reliability established separately for each downstream use.

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A Convergence and implementation verification

Table 6: Completed runs. ID and nine-heavy MAEs are independent of checkpoint selection. Wall time includes concurrent sharing of one H200.

Seed	Epochs	Best epoch	Val. MAE	ID MAE	Nine-heavy MAE	Minutes
11	1574	1474	0.2849	0.3081	0.7869	62.1
22	1226	1126	0.3398	0.3298	0.7816	51.9
33	1811	1711	0.2964	0.3121	0.7721	65.1

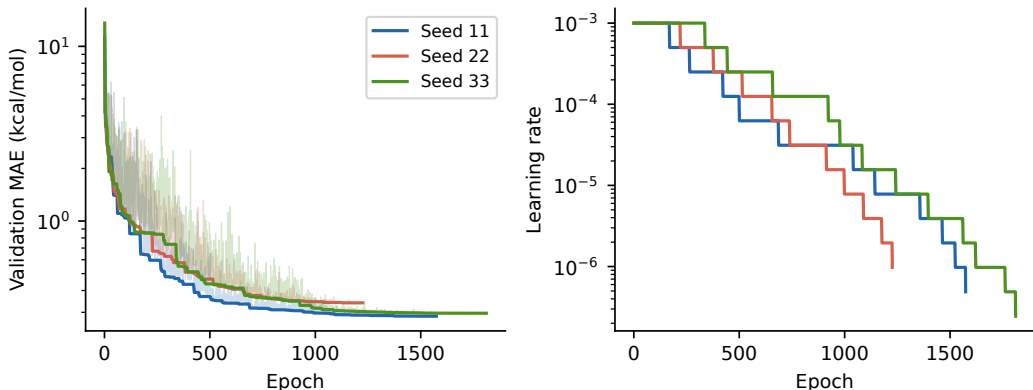


Figure 7: Complete validation histories and annealing schedules. Faint curves are epoch validation MAEs; solid curves on the left show the best value reached. Every run stops after the specified validation patience, before the 2,000-epoch ceiling.

The dense contraction reproduces the package’s sparse forward pass in a float64 engineering test: maximum differences are 5.55×10^{-16} in dimensionless energy, 3.33×10^{-16} in R , and 4.37×10^{-11} in parameter gradients. A trained-checkpoint test jointly applying a random rigid transformation and atom permutation changes molecular energy by at most 1.13×10^{-4} kcal/mol and R by 8.34×10^{-7} across 32 actual evaluation molecules. The latter tolerances reflect float32 arithmetic. Independent artifact checks reconstruct every QM9 energy from atomic orders, recompute calibration error and Holm correction, and verify disjoint heavy-atom and connectivity partitions. The corresponding records are `kernel_verification.json` and `integrity_checks.json`.

The environment uses PyTorch 2.8.0 with CUDA 12.8, NumPy 2.5.3, SciPy 1.16.0, scikit-learn 1.9.0, and RDKit 2026.03.5. The staged hippynn revision and inspected modules are documented in `codebase_notes.md`. Readout coefficients use normalized atomization-energy units during optimization; all analysis contributions are converted back to kcal/mol before dimensional continuation signals are calculated. The tiny denominator stabilizer in the implementation is 10^{-30} in squared internal energy units and does not alter ratios at the observed scales.

The companion `README.md` supplies executable reproduction commands and an artifact map. Checkpoints, every epoch’s metrics, atomic hierarchy outputs, molecular assignments, and calibrated predictions are retained. Generated numerical macros map to their source records in `paper_values.json` and `detail_values.json`; external anchors are itemized in `paper_reference.md`. Implementation corrections and negative outcomes are documented in `attempts_log.md`.

B Calibration detail and label requirements

Table 7: CCA coefficients after median normalization, fitted only on the diagnostic-fit partition. The columns follow Eq. (4). A zero is an actual constrained optimum.

Seed	Intercept	R	L	LC	T
11	0.20531	0.25681	0.00000	0.09322	0.00000
22	0.15461	0.31141	0.00000	0.06629	0.00000
33	0.13366	0.33520	0.00000	0.05803	0.00000

Table 8: Full CCA reliability data, without omitting either tail. Each entry is mean predicted / mean observed absolute error (kcal/mol). Bins contain 4,361 or 4,362 final-evaluation molecules per seed.

Bin	Seed 11	Seed 22	Seed 33
1	0.2922 / 0.2859	0.2967 / 0.3049	0.2948 / 0.2838
2	0.3619 / 0.3669	0.3858 / 0.3901	0.3561 / 0.3678
3	0.4013 / 0.4316	0.4403 / 0.4335	0.4181 / 0.4255
4	0.4746 / 0.4596	0.4718 / 0.4817	0.4850 / 0.4598
5	0.5261 / 0.5208	0.5106 / 0.5310	0.5122 / 0.5133
6	0.5748 / 0.5670	0.5647 / 0.5797	0.5661 / 0.5653
7	0.6367 / 0.6326	0.6367 / 0.6572	0.6509 / 0.6417
8	0.7439 / 0.7399	0.7501 / 0.7620	0.7178 / 0.7285
9	0.9426 / 0.9591	0.8925 / 0.9210	0.8479 / 0.8986
10	3.1717 / 2.8911	2.9627 / 2.7767	2.9217 / 2.8236

Table 9: Label-budget sensitivity for CCA. Fit and calibration sets have the stated size each; final evaluation is unchanged.

Fit labels	Calibration labels	ECE	ROC-AUC	AP
250	250	0.114 ± 0.004	0.742 ± 0.006	0.413 ± 0.014
1,000	1,000	0.074 ± 0.018	0.755 ± 0.004	0.447 ± 0.002
5,000	5,000	0.039 ± 0.001	0.756 ± 0.004	0.452 ± 0.002
43,609	21,799	0.030 ± 0.004	0.757 ± 0.004	0.461 ± 0.002

Table 10: Per-motif mean signed CCA bias ($\bar{u} - \bar{a}$, kcal/mol) and recall at $u > 1$ kcal/mol. Calibration is global and does not impose motif-conditional calibration.

Motif	Mean bias	Recall (%)
Small ring	0.057 ± 0.011	41.0 ± 2.8
Polycyclic	0.005 ± 0.016	39.9 ± 2.6
Unsaturated ring	-0.105 ± 0.009	26.3 ± 1.6
Extended conjugation	-0.091 ± 0.018	22.6 ± 0.9
Multiple carbonyls	-0.057 ± 0.013	13.6 ± 0.3
Heteroatom adjacency	-0.100 ± 0.044	33.0 ± 2.0
Heteroatom crowding	-0.068 ± 0.011	28.5 ± 1.2
Fluorinated	-0.019 ± 0.050	17.3 ± 5.0
Flexible	0.086 ± 0.007	10.7 ± 1.5
Close heteroatoms	-0.026 ± 0.023	31.3 ± 2.7
Heterocycle	-0.022 ± 0.009	33.9 ± 2.0
Triple bond	0.122 ± 0.021	19.4 ± 2.8

The fitted dimensional features use their own fit-partition medians, saved with every calibrator. Positive LC coefficients should be read conditionally on R and the chosen nonnegative regression, not as proof that cancellation physically causes error. Global isotonic calibration does not impose zero bias within each motif. The bias and recall table shows this distinction explicitly. Every threshold-sweep confusion matrix, including thresholds that produce no alerts, is included in `analysis.json`; all per-molecule predictions are retained in `uncertainty.npz`.

C Order, rank, and composition sensitivity

Table 11: Per-motif coverage of all evaluation failures and order-resolved hierarchy. $Q_n = \sum_i q_{in}$; ratios compare matching with nonmatching means. Each molecule is counted once within a row, but may match several rows.

Motif	Failure share (%)	Q_1 ratio	Q_2 ratio	Localized Δq_2 (10^{-3})
Small ring	66.0 ± 0.3	1.12 ± 0.06	2.12 ± 0.14	12.82 ± 1.07
Polycyclic	71.9 ± 0.3	1.22 ± 0.04	2.83 ± 0.12	13.69 ± 1.15
Unsaturated ring	39.8 ± 0.3	1.04 ± 0.07	0.83 ± 0.06	3.27 ± 0.82
Extended conjugation	33.0 ± 0.2	1.08 ± 0.06	0.65 ± 0.07	4.68 ± 0.99
Multiple carbonyls	4.8 ± 0.2	0.96 ± 0.05	0.52 ± 0.03	0.57 ± 0.48
Heteroatom adjacency	14.1 ± 0.3	1.28 ± 0.08	0.94 ± 0.20	6.70 ± 3.66
Heteroatom crowding	46.1 ± 0.3	1.05 ± 0.01	0.91 ± 0.10	4.28 ± 0.87
Fluorinated	1.6 ± 0.1	0.98 ± 0.08	0.67 ± 0.10	2.50 ± 2.24
Flexible	2.8 ± 0.1	0.75 ± 0.03	0.37 ± 0.01	2.35 ± 0.29
Close heteroatoms	14.6 ± 0.2	1.03 ± 0.02	0.93 ± 0.04	4.60 ± 1.55
Heterocycle	85.4 ± 0.2	1.34 ± 0.07	1.49 ± 0.10	9.12 ± 1.12
Triple bond	13.36 ± 0.05	1.01 ± 0.05	0.52 ± 0.02	1.32 ± 0.46

Table 12: Descriptive rank effect sizes underlying the confirmatory QM9 tests. Each probability is the Mann–Whitney statistic divided by the number of matching/nonmatching graph pairs, with half weight for ties. A value above 0.5 indicates larger ranks, which need not imply a larger arithmetic mean.

Motif	$P(a_+ > a_-)$	$P(R_+ > R_-)$	Median R_+/R_-
Small ring	0.527 ± 0.002	0.591 ± 0.016	1.13 ± 0.03
Polycyclic	0.620 ± 0.003	0.698 ± 0.011	1.32 ± 0.02
Unsaturated ring	0.563 ± 0.002	0.549 ± 0.017	1.07 ± 0.03
Extended conjugation	0.544 ± 0.001	0.562 ± 0.028	1.10 ± 0.05
Multiple carbonyls	0.506 ± 0.005	0.456 ± 0.013	0.95 ± 0.02
Heteroatom adjacency	0.577 ± 0.005	0.642 ± 0.039	1.30 ± 0.12
Heteroatom crowding	0.5564 ± 0.0004	0.569 ± 0.010	1.10 ± 0.02
Fluorinated	0.522 ± 0.007	0.513 ± 0.032	1.04 ± 0.06
Flexible	0.349 ± 0.004	0.248 ± 0.004	0.71 ± 0.01
Close heteroatoms	0.528 ± 0.003	0.532 ± 0.003	1.047 ± 0.004
Heterocycle	0.609 ± 0.002	0.691 ± 0.005	1.33 ± 0.02
Triple bond	0.411 ± 0.003	0.443 ± 0.029	0.94 ± 0.05

Table 13: Exploratory exact-formula standardization. Only formulas with at least five matching and five nonmatching molecules contribute; the two group means use identical formula weights. These descriptive ratios are not new significance tests.

Motif	Formulas	N_+	N_-	Error ratio	R ratio
Small ring	75	27,270	13,960	1.49 ± 0.01	1.72 ± 0.03
Polycyclic	71	23,123	15,738	2.11 ± 0.02	1.97 ± 0.04
Unsaturated ring	67	12,254	26,500	0.87 ± 0.02	0.72 ± 0.01
Extended conjugation	66	10,108	27,090	0.66 ± 0.02	0.61 ± 0.01
Multiple carbonyls	36	2,097	15,725	0.739 ± 0.004	0.66 ± 0.02
Heteroatom adjacency	60	3,605	20,544	0.92 ± 0.01	1.00 ± 0.08
Heteroatom crowding	66	14,292	18,271	1.15 ± 0.01	1.03 ± 0.02
Fluorinated	0	0	0	not estimable	not estimable
Flexible	60	4,316	32,618	0.479 ± 0.003	0.57 ± 0.03
Close heteroatoms	78	5,422	28,248	1.235 ± 0.002	1.28 ± 0.01
Heterocycle	73	29,593	10,257	1.74 ± 0.02	1.54 ± 0.03
Triple bond	66	10,381	28,609	0.34 ± 0.01	0.48 ± 0.01

The primary confirmatory family concerns joint upward ranks in error and R , whereas discovery selection for the Pareto curves uses mean directions. These operations are deliberately distinct. No significance conclusion is based on the exploratory exact-formula or intersection tables. Fluorination has no within-formula contrast because its membership is fixed by elemental composition; the corresponding entry is not estimable, rather than zero. Formula-standardized summaries use only

Table 14: Exploratory intersections of the three ring-related candidates. S: small ring; P: polycyclic; H: heterocycle. Every final-evaluation molecule belongs to exactly one row. This disjoint accounting does not redefine the candidate library.

Exact membership	N	MAE	Failure rate (%)	Share of failures (%)
None	5,351	0.419 ± 0.003	6.4 ± 0.2	4.3 ± 0.1
S	2,770	0.360 ± 0.004	4.6 ± 0.1	1.6 ± 0.1
P	172	0.678 ± 0.032	22.1 ± 3.1	0.5 ± 0.1
S+P	4,043	0.727 ± 0.005	15.9 ± 0.3	8.1 ± 0.1
H	8,921	0.705 ± 0.006	17.5 ± 0.3	19.8 ± 0.3
S+H	3,399	0.392 ± 0.003	5.5 ± 0.3	2.4 ± 0.1
P+H	1,791	1.254 ± 0.030	41.5 ± 0.9	9.4 ± 0.2
S+P+H	17,169	1.040 ± 0.012	24.8 ± 0.5	53.8 ± 0.3

the supported strata and need not reproduce the population-wide association. Counts and the full unrounded statistics are in `descriptive_checks.json`.

D Size-resolved conformational transfer

Table 15: ANI relative-energy discrepancy by molecular size. Rates condition on evaluated conformers. The model was trained on optimized geometries with at most eight heavy atoms.

Heavy atoms	Graphs	Conformers	Mean discrepancy	Rate > 1 (%)
≤ 8	1,461	8,029	19.44 ± 0.20	95.8 ± 0.2
9	27	175	24.83 ± 0.41	97.1 ± 0.3
≥ 10	231	1,040	22.65 ± 0.54	96.1 ± 0.2

Table 16: Size-stratified ANI motif effects. Each cell gives matching / nonmatching graph counts and the ratio of their mean discrepancies, averaged across seeds. These are descriptive conditional effects, without additional hypothesis tests.

Motif	≤ 8 heavy atoms	9 heavy atoms	≥ 10 heavy atoms
	311 / 1150	2 / 25	6 / 225
Small ring	1.31 ± 0.02	1.08 ± 0.03	1.35 ± 0.05
	194 / 1267	0 / 27	59 / 172
Polycyclic	1.59 ± 0.01	not estimable	1.14 ± 0.04
	670 / 791	11 / 16	107 / 124
Unsaturated ring	0.99 ± 0.01	0.86 ± 0.01	1.11 ± 0.02
	906 / 555	19 / 8	216 / 15
Extended conjugation	0.94 ± 0.02	0.79 ± 0.03	0.55 ± 0.02
	53 / 1408	7 / 20	158 / 73
Multiple carbonyls	0.74 ± 0.01	0.68 ± 0.02	0.71 ± 0.02
	754 / 707	8 / 19	21 / 210
Heteroatom adjacency	1.04 ± 0.02	1.98 ± 0.06	1.33 ± 0.03
	883 / 578	24 / 3	202 / 29
Heteroatom crowding	1.02 ± 0.02	0.96 ± 0.02	0.61 ± 0.01
	0 / 1461	0 / 27	0 / 231
Fluorinated	not estimable	not estimable	not estimable
	39 / 1422	10 / 17	173 / 58
Flexible	1.06 ± 0.02	1.02 ± 0.03	0.73 ± 0.03
	369 / 1092	14 / 13	182 / 49
Close heteroatoms	1.17 ± 0.01	0.86 ± 0.01	0.84 ± 0.02
	857 / 604	12 / 15	97 / 134
Heterocycle	1.17 ± 0.01	0.908 ± 0.004	1.16 ± 0.03
	150 / 1311	0 / 27	3 / 228
Triple bond	0.840 ± 0.003	not estimable	2.11 ± 0.07

The sampling manifest retains every source HDF5 bucket/row and every inferred connectivity. The offset and evaluation halves are disjoint within each eligible graph. Motif transfer significance uses graph-level mean discrepancies, while the reported rates condition on conformers; neither conformers of a molecule nor repeated fits are pooled as independent hypothesis-test replicates. A variable close-contact assignment is summarized by majority exposure for graph-level tests. Exact assignment fractions remain available in the sampled-data artifact.